Status	Finished
Started	Wednesday, 21 October 2026, 6:12 PM
Completed	Wednesday, 21 October 2026, 6:23 PM
Duration	11 mins 28 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Backtracking
b. Greedy method
c. Dynamic programming
d. Divide and conquer

The correct answer is:
Divide and conquer

Question **2**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It bubbles up the largest element in each pass
b. It divides the list into sublists
c. It uses the heap data structure
d. It recursively sorts subproblems

The correct answer is: It bubbles up the largest element in each pass

Question **3**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It simplifies the sorting process by using only one pass
b. It avoids the need for recursion
c. It only works on small datasets
d. It allows for efficient parallel processing and sorting of data

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **4**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Bubble Sort
b. Binary Search
c. Linear Search
d. Merge Sort

The correct answer is: Bubble Sort

Question **5**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort is less efficient for large lists
b. Bubble Sort is difficult to implement
c. Bubble Sort cannot handle duplicate elements
d. Bubble Sort does not guarantee sorted order

The correct answer is: Bubble Sort is less efficient for large lists

Question **6**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It sorts data using a single pass
b. It is less efficient than bubble sort
c. It is based on the divide-and-conquer approach
d. It does not require recursion

The correct answer is: It is based on the divide-and-conquer approach

Question **7**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Combining sorted sublists
b. Dividing the input into smaller subproblems
c. Comparing each element with the others
d. Sorting the entire list sequentially

The correct answer is: Dividing the input into smaller subproblems

Question **8**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Quick Sort
b. Bubble Sort
c. Merge Sort
d. Insertion Sort

The correct answer is: Bubble Sort

Question **9**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

- a. It makes it easier to select items based on their relationship to the rest of the items
- b. It slows down the process
- c. It complicates the selection of items
- d. It makes the data unsorted

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **10**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Merge Sort
- b. Linear Search
- c. Quick Sort
- d. Bubble Sort

The correct answer is: Merge Sort

Question **11**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Makes data harder to manage
- b. Decreases the efficiency of algorithms
- c. Makes code execution slower
- d. Provides a basis for other algorithms to work efficiently

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **12**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are discarded
b. They are combined to form a single sorted list
c. They are split again into smaller sublists
d. They are compared element by element

The correct answer is: They are combined to form a single sorted list

Question **13**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. sorted()
b. arrange()
c. order()
d. sort()

The correct answer is: sort()

Question **14**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Quick Sort
b. Merge Sort
c. Insertion Sort
d. Bubble Sort

The correct answer is: Merge Sort

Question **15**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

- a. It allows for faster searching
- b. It has no effect on the search operation
- c. It makes the search operation slower
- d. It increases the number of comparisons needed

The correct answer is: It allows for faster searching